ahmad_mahmoud.resume.ts

FULL-STACK ENGINEER ● 12+ YRS

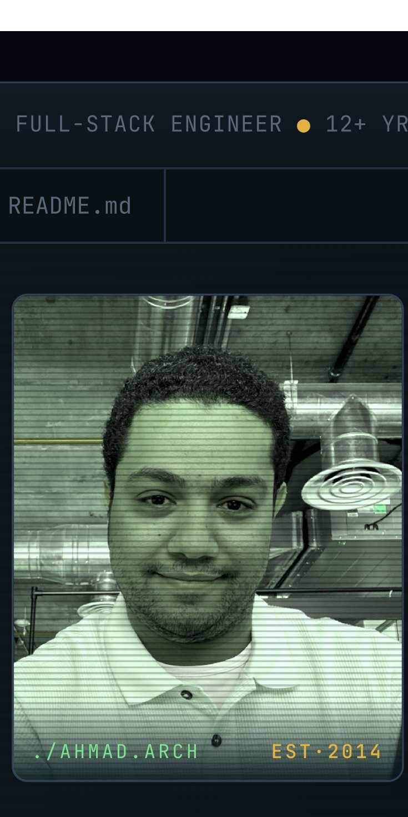
● full_stack.ts | experience.json | projects/ | stack.config | README.md

→ whoami --full

Ahmad Abdul-Jawad Mahmoud

Full-Stack Engineer / Founder & CTO, Buraq

// **backend** · **web** · **mobile** · **12+ years** in production · Cairo ⟶ Riyadh · database to App Store



9	330+	2,900+	12+
REPOS BY ONE AUTHOR	API ENDPOINTS	BACKEND TESTS	YEARS ENGINEERING

```
/**
* Full-Stack Engineer — I own a feature from the database to the
* app on the phone. Since March 2026: Founder & CTO of Buraq (buraqegypt.com),
* an Egyptian ride-hailing platform built solo across every layer — a NestJS /
* TypeScript backend on PostgreSQL and Redis (22 modules, 330+ endpoints,
* 2,900+ tests), a 29-screen ops console, the public web app (live), realtime
* Socket.IO feeds, payments, and native iOS & Android apps on one API contract.
* Behind it: 12+ years of production engineering, seven of them owning a
* POS used by 20k+ restaurants.
*/
```

## across_the_stack()

4 pillars

[01] backend & data

Correct before clever

A NestJS modular monolith (**22 modules**) on PostgreSQL with versioned migrations; Redis for dispatch state, the driver geo index, rate limits and cron leases. A **double-entry ledger**, a transactional outbox and idempotency keys, so money and side effects survive retries.

◆ 22 modules · ledger · outbox

[02] realtime & integrations

The parts that talk to the world

Socket.IO gateways for riders, drivers and admins; Google Maps routing with traffic-aware fare quotes and caching; Paymob cards, Apple Pay and Google Pay; WhatsApp-first OTP with SMS fallback; APNs and FCM push.

◆ Sockets · maps · payments · OTP · push

[03] web

An ops console people can run a business on

A **29-screen** Next.js 14 console: live MapLibre trips map, dispatch and SOS boards, KYC review, finance reconciliation, refunds and payouts, a map zone editor, feature flags, **RBAC** and an audit log behind Google SSO.

◆ 29 screens · RBAC · audit log

[04] mobile clients

12 years of shipping apps

Native SwiftUI (Swift 6) and Kotlin / Compose apps on shared, versioned core libraries, with **1,000+ iOS** and **4,000+ Android tests**. Before this: apps used by **35M+ people** — HungerStation, Ego ride-hailing and the Marn POS.

◆ SwiftUI · Compose · 5,000+ client tests

## experience()

8 roles

Founder & CTO — Backend · Web · iOS · Android @ Buraq — buraqegypt.com

Mar 2026 → Present CURRENT

CAIRO · FOUNDER · BURAQ TECHNOLOGIES

Founded an Egyptian ride-hailing platform and have carried its **entire technical scope alone**: chose the stack, set the standards, and built eleven repositories from empty — a hardened backend and operations console, rider and captain apps on **iOS and Android**, a shared core package per platform, and the public web app **live at buraqegypt.com**. Every layer governed by explicit rules: idempotent mutations, double-entry accounting, acyclic module boundaries, and a CI suite that gates the merge. Now preparing a cash-only lean beta in Cairo.

› Backend API
// NestJS · TypeScript · Mar '26 → now

Modular monolith, **22 modules** in an acyclic graph — **trip state machine**, **scored waterfall dispatch** over a Redis geo index, admin-tunable pricing across six ride tiers, **double-entry ledger** with cash and card settlement (Paymob, Fawry), scheduled and multi-stop rides, five realtime **Socket.IO gateways**, OTP / Apple / Google auth with RBAC, idempotency and audit log · **330+ endpoints** over 55 controllers, 62 entities, **2,900+ tests**, Swagger docs.

› Ops Console
// Next.js 14 · Jul '26 → now

29-screen ops surface — live **MapLibre** trips map with demand heatmap, dispatch and SOS boards, KYC review, finance reconciliation, refunds and payouts, cash control, map **zone editor**, feature flags, audit log, **Google SSO** · English and Arabic, 170+ tests.

› iOS · Rider & Captain
// SwiftUI · Swift 6 · Jul '26 → now

Two native **SwiftUI** apps on a shared **BuraqCore** package (domain · networking · design system, version-pinned) — the full **trip lifecycle** on live GPS, Socket.IO feeds with fresh-token reconnect, chat, wallet and payments, **Live Activities**, Siri voice booking, on-device ID verification, captain earnings and navigation · ~240 views, **1,000+ tests**, strict concurrency.

› Android · Captain
// Kotlin · Jetpack Compose · Aug '26 → now

Native port against **the same API contract** — 18 feature modules on convention plugins, Hilt, a pure-Kotlin domain; onboarding, the shift's foreground service and the trip end to end running against the backend · **1,500+ tests**.

› Android · Rider
// Kotlin · Jetpack Compose · Sep '26 → now

Screen-by-screen parity port of the iOS rider app — 11 feature modules on a published **buraq-core-android** (GitHub Packages) · **2,200+ tests**, including 371 screenshot goldens in light, dark and RTL.

› Web · buraqegypt.com
// Next.js 16 · React 19 · Oct '26 → now

Marketing site, **rider web app** with live trip tracking, **captain onboarding portal** with on-device document OCR, public trip-share pages and app links · English and Arabic (RTL), WCAG 2.2 AA, **1,800+ tests** · **live on Render since 15 Oct 2026**.

› iOS · Buraq Schools
// SwiftUI · Swift 6 · Nov '26 → now

A second product on the same platform: **one app for parents and school-bus supervisors** — the live bus, safe handovers, an offline trip mode and an **iOS 26 Liquid Glass** interface · **1,000+ tests**; the Android port is under way.

One backend · one contract · eight client surfaces — and the company around them: incorporation, the Law 87/2018 licence path, a pricing engine fitted to 626 observed fares, the business model and investor materials, the go-live plan. 3,100+ commits, one author — so I know exactly what I will be asking of the team I hire.

iOS Team Lead @ Marn — Jahez Group

Aug 2019 → Aug 2026

RIYADH, KSA

- **Led a 5+ engineer iOS team** for seven years — onboarding, mentoring, career growth, and the review culture behind an enterprise POS used by **20k+ restaurants** and cafés.
- **Owned the technical roadmap and architecture:** an offline-first CoreData core with cloud sync, built so a venue keeps trading when the network does not.
- **Set the engineering standards** — module boundaries, code review, automated testing — holding the app at a **99.8–99.9% crash-free** rate in production.
- **Drove the multi-year migration** to SwiftUI, Combine and async/await without pausing feature delivery, and made the case for it to product leadership.
- **Built CI/CD and the release process from scratch**, cutting deployment time **60%** and shortening every engineer's feedback loop.
- Ran the performance programme (**40% faster load times**) and designed the table-management, order-processing and inventory systems at the product's core.
- Partnered with product, design and backend leadership on scope, estimates and delivery commitments.

Senior iOS Engineer @ HaramBlur · voluntary charity project

Dec 2025 → Jun 2026

- Designed an architecture integrating **on-device machine learning** for real-time content analysis.
- Built a seamless communication layer between native iOS and web-based content.
- Implemented efficient image processing with region-based and full-image blur effects.
- Created a real-time video monitoring system for continuous content filtering.
- Developed smart banner blocking and optimised memory management throughout.

iOS Team Lead @ Ego — Ride Hailing

Mar 2018 → May 2019

RIYADH, KSA

- Developed a ride-hailing ecosystem (iOS, watchOS) serving **15M+ users** in Saudi Arabia.
- Implemented real-time GPS tracking and navigation with MapKit and optimised route algorithms.
- Integrated Apple Pay, HyperPay, and push-notification infrastructure.
- Created seamless iPhone ⟷ Apple Watch handoff for continuous ride tracking.
- Built the driver rating system and real-time fare calculation engine.

Senior iOS Engineer @ HungerStation — DeliveryHero

Sep 2016 → Mar 2018

KSA

- Contributed to a major food-delivery platform serving **20M+ users**.
- Optimised app performance for high-volume transaction processing.
- Implemented real-time order tracking and delivery optimisation.
- Collaborated with backend teams on API design and integration.

iOS Developer @ ALNapedh

May 2016 → Aug 2016

- Developed food-ordering apps with Firebase integration and real-time features.
- Implemented Spreedly payments and live-chat functionality.
- Built responsive UI with Auto Layout across multiple device sizes.

iOS & Apple Watch Developer @ Nile Design Co.

Jul 2015 → Sep 2015

- Early adopter of Apple Watch development (watchOS 1–2).
- Built a foundation in iOS and watchOS app development.
- Adopted SwiftUI from WWDC 2019 onward.

iOS Developer @ UK Pro. Solutions Ltd.

May 2014 → May 2015

- Foundation iOS experience in Objective-C and early Swift.
- Built social networks and e-commerce platforms.
- Learned core iOS principles and App Store deployment.

## projects[]

selected

[01] ai / privacy

HaramBlur — Privacy Browser

SwiftUI · CoreML · WKWebView · CoreImage · Vision

Privacy-focused iOS browser with real-time, AI-powered content detection and filtering. On-device ML, smart blur, live video monitoring.

◆ AI-powered privacy browser

[02] full-stack

Buraq — Ride-Hailing

NestJS · TypeScript · PostgreSQL · Redis · Socket.IO · Next.js · SwiftUI · Kotlin · Jetpack Compose

Egyptian ride-hailing platform I founded and built solo: a hardened NestJS backend (trip state machine, scored dispatch, double-entry ledger, five realtime gateways, 2,900+ tests), a 29-screen ops console, rider and captain apps on **iOS and Android**, and the public web app **live at buraqegypt.com**.

◆ Founder & CTO · nine repos · 10k+ tests

[03] delivery

HungerStation

UIKit · Firebase · Real-time APIs · Perf

Major food-delivery platform serving 20M+ users across Saudi Arabia and Bahrain. Performance and real-time order tracking.

★ 4.5 / 5 — 1.9M+ ratings

[04] mobility

Ego — Ride Hailing

MapKit · Core Location · Apple Pay · watchOS

Complete ride-hailing ecosystem: rider, driver, and Apple Watch apps. Real-time GPS and cross-device experiences for 15M+ users.

★ 4.7 / 5 — 211.8k+ ratings

[05] enterprise

Marn — iPad POS

SwiftUI · CoreData · CloudKit · Real-time Sync

Enterprise POS serving 20k+ restaurants across Saudi Arabia. Offline-first architecture, complex table management, real-time orders.

★ 4.8 / 5 — 2,437 ratings

[06] community

Daleelo

MapKit · Core Location · Social SDKs

Muslim community app — prayer times, mosque finder, and community features with enhanced location + social integration.

★ 4.5 / 5 — 1,287 ratings

## stack.config

web_and_infrastructure =
Next.js 14 · React · TypeScript · MapLibre · Socket.IO · Docker · GitHub Actions CI · Prometheus Metrics · Jest · Google Maps Platform · Paymob · Cloudflare R2

ios_2025 =
Swift 6.0 · SwiftUI · UIKit · Combine · async/await · SwiftData · iOS 18+

android_2025 =
Kotlin · Jetpack Compose · Android · Coroutines · Flow · MVVM

architecture =
MVVM · VIPER · Clean Architecture · Coordinator · TDD

apple_ecosystem =
watchOS · tvOS · macOS · WidgetKit · Live Activities · App Intents

intelligence =
Core ML · Vision · ARKit · Dynamic Island · App Tracking

backend_apis =
NestJS · TypeScript · PostgreSQL · TypeORM · Redis · JWT/OTP · REST · GraphQL · Firebase · CloudKit · Socket.IO · Swagger

tooling =
Xcode 15+ · Fastlane · TestFlight · Instruments · Git · Claude Code · Copilot · Cursor

## portfolio/

FOOD & GAMING
- HungerStation
- Punters

TRANSPORTATION
- Buraq — founder
- Ego — Rider
- Ego — Driver
- Ego — Watch

ENTERPRISE
- Marn POS
- Medical Guide

SOCIAL & COMMUNITY
- Daleelo
- NU-inc
- VeeU

WATCHOS
- Ego Watch
- TaQs Weather
- TweetMe

E-COMMERCE
- iBss
- Exclusive Luxe
- Scratch

RELIGIOUS & EDU
- Afasy
- ArabNileGroup

UTILITY & TOOLS
- Unlocking Smart
- Insta
- iContacts
- First iOS app

AI & ML
- HaramBlur
- RecognizeMe
- Grad project · OpenCV

WEB
- FaceDemo
- Web Pages

## record{}

// education

B.Sc. Information Systems
Computer Sciences College
Asyut University, Egypt · 2009–2013
iOS Semester — Excellent (91.5%)

// certification

ISTQB Foundation Level
Egyptian Software Testing Qualification Board

// languages

Arabic	native
English	professional

## honours[]

01 Building **Buraq solo across every layer** — backend, ops console, web app, and iOS and Android apps on one API contract.

02 Held a **99.8–99.9% crash-free** rate and cut deployment time **60%** with the standards, tests and CI/CD the team ran on.

03 **Founded Buraq** (buraqegypt.com), an Egyptian ride-hailing platform, and built all eleven repositories solo — backend, ops console, rider and captain apps on iOS and Android, Buraq Schools, and the public web app, live since Oct 2026 — behind **10,000+ automated tests**.

04 Built a food-delivery platform serving **20M+ users** and a ride-hailing ecosystem serving **15M+ users** in Saudi Arabia.

05 **10+ apps** on the App Store, used by millions across MENA.

06 **30+ projects** delivered across a 12-year career.

07 Early Swift adopter (WWDC 2014) & early SwiftUI adopter (WWDC 2019).

main | ● Full-Stack Engineer · Backend · Web · Mobile | Cairo ⟶ Riyadh | UTF-8 | TypeScript · Swift · Kotlin | ★ 12+ yrs · founder & CTO, Buraq